

Neighbor balance and evenness of distribution of treatment replications in row-column designs

Hans-Peter Piepho¹  | Volker Michel² | Emlyn Williams³

¹Biostatistics Unit, University of Hohenheim, Stuttgart, Germany

²Landesforschungsanstalt für Landwirtschaft und Fischerei Mecklenburg-Vorpommern, Gülzow, Germany

³Statistical Consulting Unit, Australian National University, Canberra, Australia

Correspondence

Hans-Peter Piepho, Biostatistics Unit, University of Hohenheim, Fruwirthstrasse 23, 70593 Stuttgart, Germany.

Email: piepho@uni-hohenheim.de

Abstract

Row-column designs allow error control in field experiments by blocking in two dimensions. While this strategy can capture spatial heterogeneity aligned with blocks and account for effects due to the farming operations along rows and columns, it suffers from the occasional clustered occurrence of several replications of the same treatment. This property of classical row-column designs has hampered their more widespread use in practice. A further issue of practical importance is the degree of neighbor balance of a design, that is, the frequency of adjacencies of pairs of treatments. This paper proposes two design strategies to simultaneously optimize the evenness of spatial distribution of treatment replication as well as neighbor balance. Three examples are given to illustrate the proposed methods and demonstrate that both approaches yield comparable and satisfactory results.

KEYWORDS

average efficiency factor, comparative experiment, experimental design, gerechte designs, Latinization, t-Latinization, superblocks

1 | INTRODUCTION

Many field experiments are laid out on a rectangular grid of plots. The two-dimensional layout provides the opportunity to control for environmental gradients in both row and column directions by blocking. Efficient row-column designs are readily available using combinations of theory and computer-aided search strategies (John & Williams, 1995), but in our experience they are still underutilized. Many experimenters design their trials expecting row and column effects to be present, but use complete block designs having blocks aligned with only the rows or only the columns. To cater for any effects in the direction orthogonal to the blocks they then manually restrict randomization within blocks so that no treatment appears more than once in a row or column of the layout. One common way of doing this in practice is to take a randomized layout for a randomized complete block design (RCBD) and manually swap treatments within blocks until a desired design is obtained. Another approach is to allocate treatments to blocks manually so that a seemingly desirable pattern is obtained. The fact that such ad hoc procedures imposing additional randomization restrictions on columns are often used, is clear indication that row-column designs are urgently needed in many field experiments. Despite this need, however, randomized row-column designs as yet seem to be used relatively rarely in field experiments.

In our experience the use of classical row-column designs has been hampered by the fact that randomization may occasionally result in a clustered occurrence of replications of some of the treatments in a restricted area of the experimental field (Piepho, Michel, & Williams, 2016). Such clustering is deemed undesirable by experimenters who are concerned that irregular environmental gradients may cause several replications of the same treatment to be adversely affected, thus potentially causing bias in treatment effect estimates. Hence, there is a felt need for a design strategy that avoids clustering and distributes treatment replications evenly over the experimental field. We think that the ad hoc modifications so often used in practice target two desirable

design properties that will be detailed below. The main purpose of this paper is to provide methods for generating row-column designs that have these two desirable properties, thus fostering the more common use of row-column designs in applied research.

The first property targeted by our approach is an even distribution of treatment replications (henceforth abbreviated as ED). A simple option to achieve ED is to employ a resolvable design, that is, a design where rows and columns can be grouped to form complete replicates (John & Williams, 1995). Furthermore, with resolvable designs, Latinization can be used as an additional strategy to evenly distribute treatment replications (John & Williams, 2002). Latinization involves grouping rows and/or columns of different replicates to form new blocks extending across several replicates. Ensuring that each treatment occurs at most once in a replicate and in a new block formed by Latinization leads to improved ED across the trial (Piepho, Michel, & Williams, 2015). There are restrictions on the dimensions of replicates and Latinized blocks in resolvable designs, however, and these dimensions may not be optimal to capture a substantial share of the environmental variation. Also, in many applications, resolvability is not a feasible option due to constraints on the number of rows and columns as well as the number of treatments of the design. For example, in postregistration variety trials in Germany, trials are typically laid out in very few rows, with the number of columns exceeding that of rows and with plots that are longish in column direction. In many cases, rows correspond to complete blocks, that is, the number of columns equals the number of treatments. In this setting the number of treatments to be tested may not permit a resolvable row-column design to be used. There are also instances where the number of columns is smaller than the number of treatments. Moreover, in some cases, the number of replications is not the same for all treatments. This may happen, for example, when the field layout has been determined for an equally replicated design, but then on short notice treatments are added to, or dropped from the original design. In such instances, the total number of plots may not be a multiple of the number of treatments, meaning that different replication numbers will need to be used for the various treatments. Similarly, the experimenter may choose to have extra replication for some of the treatments because of their special importance or because they provide information on the error variance and block effects, whereas most of the treatments remain unreplicated; augmented designs (Federer, Nair, & Raghavarao, 1975) and partially replicated designs (Smith, Lim, & Cullis, 2006; Williams, Piepho, & Whitaker, 2011) are pertinent examples of the latter. In summary, there is a need to devise a design strategy for such nonresolvable and potentially unequally replicated settings that ensures ED across the experimental field.

Apart from ED, a second and related consideration in the layout of blocked field experiments is neighbor balance (Azais, Bailey, & Monod, 1993; Rees, 1967), subsequently abbreviated as NB. Complete NB occurs when each treatment pair appears on adjacent plots the same number of times. This condition is also known as first-order NB (Wilkinson, Eckert, Hancock, & Mayo, 1983; Williams, 1952). In many experiments, the dimensions of the field layout and the number of treatments do not permit complete NB, but one may strive for partial NB, meaning that the number of adjacencies does not differ by more than one among all treatment pairs (Chan & Eccleston, 1998). Many proposals have been made for (partially) neighbor-balanced designs with one-dimensional blocking and software has recently been made available to generate randomized layouts from published catalogues (Jaggi, Varghese, Varghese, & Sharma, 2015). Extensions to two-dimensional blocking have also been proposed (Chan & Eccleston, 1998, 2003; Varghese, Jaggi, & Varghese, 2014). Here, we are focusing exclusively on NB along the rows because in the trials we consider, plots are longish down the columns, and there is space left for driveways between rows of plots, meaning that neighbor effects are expected in the row direction only.

One important reason to strive for some degree of NB is that occasionally all replications of a treatment may be lost completely, for example, due to frost damage, leaving essentially bare plots after winter. It is common practice in such cases to also drop all neighboring plots in the same row from statistical analysis, because they are adversely affected by neighbor effects. If the lost treatment has many adjacencies with a particular other treatment, that other treatment will also suffer a substantial loss of replications. Thus, complete or partial NB ensures that the dataset remaining after removal of the lost treatment and its neighboring plots maintains a certain degree of balance for the remaining treatments.

For nonresolvable row-column designs, Piepho et al. (2016) recently proposed a design strategy that avoids clustering, that is, achieves good ED, but observed that at the same time the generated designs often displayed some notable imbalance regarding treatment adjacencies. It appears from these results that there is a conflict between the contrasting objectives of ED and NB along the rows. The present paper therefore proposes two design strategies that can address the objectives of both ED and NB simultaneously. The strategies are based on a computer-aided design search. A key feature is the formation of blocks that are superimposed onto the row-column structure. This can be viewed as a generalization of the process of Latinization, which has been denoted as *t*-Latinization (John & Williams, 2002). We present two approaches. The first approach is based on a linear mixed model with spatial covariance (Eccleston & Chan, 1998). We show how emphasis can be focussed on either of the two objectives (ED and NB) by tuning the variance parameters of an assumed spatial variance-covariance matrix for the residual plot errors. The second approach directly optimizes a set of criteria that assess features related to ED and NB. Our exposition starts with a motivating example from a recent variety trial. We then describe the two approaches and illustrate them using the motivating example. Further examples serve to demonstrate the versatility of the two approaches.

20	18	23	13	7	9	12	16	11	19	1
5	24	22	19	1	14	3	9	18	23	10
22	20	12	21	14	10	8	18	13	17	4
8	15	10	11	16	19	7	4	23	14	25
3	16	21	24	25	6	14	22	15	13	20
1	19	6	7	2	17	18	25	24	21	8
16	5	1	4	10	11	6	17	3	12	24
11	12	15	22	13	2	5	6	8	25	9
21	2	7	9	3	20	4	23	17	5	15

FIGURE 1 Two-dimensional layout for $v = 25$ treatments, $k = 9$ rows, and $s = 11$ columns $[(v, k, s) = (25, 9, 11)]$ generated using model-based approach with $(\gamma_R, \gamma_C, \gamma_E) = (10^3, 10^3, 1)$. The following statistics are explained later in the paper: $\bar{f}_A^{RC} = 0.864$; $(n_0, n_1, n_2, n_3) = (222, 67, 10, 1)$; $MST_2 = 3.17$ (treatment 2 is the treatment with only three replications), $MST_6 = 2.27$ (smallest among treatments with four replications)

2 | A MOTIVATING EXAMPLE

Example 1: The ‘Landesforschungsanstalt für Landwirtschaft und Fischerei’ (LFA) in the federal state of Mecklenburg-Vorpommern, Germany, conducts an EU variety series of trials with forage maize commissioned by the ‘Pro-Corn Gesellschaft für Beratung und Sortenprüfung mbH’ (Pro-Corn). The results of this series of trials are also used for deriving variety recommendations. The trials were originally planned to have $r = 4$ replications at each location for each of $v = 23$ varieties, requiring a total of 92 plots. In the past, most experimenters conducting the trials for this series have used a randomized complete block design with four replicates laid out in four rows and 23 columns, with rows corresponding to complete blocks.

In order to allow blocking in two dimensions, the LFA formerly would have employed a resolvable row-column design for $v = 24$ treatments, for example, with $k = 8$ rows and $s = 12$ columns, filling the plots for the extra treatment with a further variety and using Latinization to ensure even distribution of treatment replications. Complete replicates would have been formed from 2-by-12 or 3-by-8 or 4-by-6 row-by-column grids of plots. The space available for one particular trial, however, allowed for $k = 9$ rows and $s = 11$ columns, which did not fit the requirements for a resolvable design with $v = 23$ varieties. Thus, a total of 99 plots were available rather than only 96 plots for the resolvable design and the 92 plots needed to accommodate 23 varieties with four replications each. Eventually, it was decided to use all of these 99 plots to design a trial for $v = 25$ varieties, one of which would only have three instead of four replications. Obviously, no resolvable design is available in this situation.

To set the stage, we consider a classical row-column design generated for this setting (Piepho et al., 2015). The design is shown in Figure 1. Information given in the figure caption refers to the details of the method and descriptive measures described in the next section. Each row and each column forms a binary block, that is, no treatment occurs more than once in a column or row. There are, however, 10 treatment pairs with two adjacencies and one pair (1, 19) with three adjacencies in rows, so there is scope for improving NB. Also, there is one diagonal self-adjacency for treatments 6 and 18, and the four replications of treatment 6 are relatively close to each other, so there is some degree of clustering. Thus, to an experimenter the ED of treatment replications may not seem fully satisfactory either, although we should stress that the design is perfectly valid. Nevertheless, many research scientists will reject such a design as “suboptimal” due to poor NB or ED or both and consider an alternative ad hoc modification. The design strategies described in the next section will therefore be used subsequently to improve ED as well as NB.

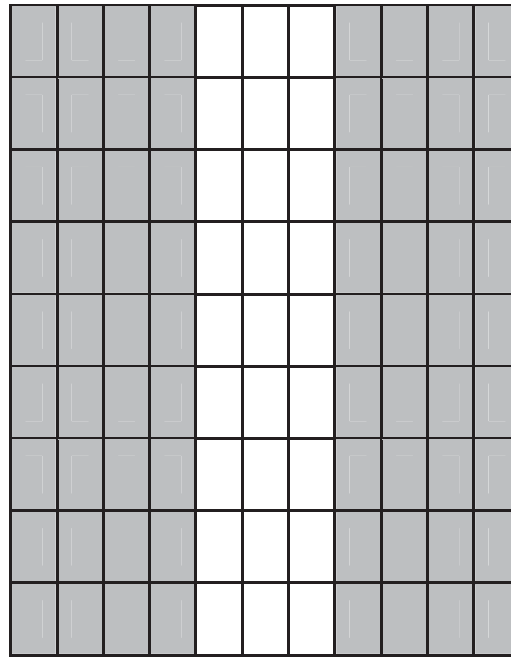


FIGURE 2 A layout with $k = 9$ rows, $s = 11$ columns, and $g_s = 3$ rectangular columngroups formed by grouping four, three, and four columns

3 | THE GENERAL APPROACH

We here present two alternative design generation methods, the one based on a linear-mixed model, the other on a direct optimization of NB- and ED-related features of the design. A common key idea of both approaches is to impose superblock structures onto the row-column field layout in order to achieve better ED. In our model-based approach, the overall blocking structure is represented by a mixed model with random block effects. Spatial covariance structures are imposed for the residual plot errors in order to improve NB and further improve ED. The resulting variance-covariance structure is then used in a numerical search for good designs. In our direct optimization approach, the block structure is maintained throughout the numerical search without explicitly using a model and treatment composition of blocks is modified consecutively by swap operations. We here first discuss the superblock structures imposed in order to achieve ED across the experimental area. Then we describe the criteria used to evaluate the designs. Next, the linear mixed models that are used for the model-based design optimization regarding ED and NB are described, followed by the direct optimization method.

3.1 | Block structures

The basic block units for our designs are the rows and columns of the field layout. We are assuming that the long sides of the plots run down the columns so it will generally be desirable to have $k \leq s$ in order that the shape of the trial area is as near to a square as possible. Rows and columns may be overlaid with superblocs, which we denote as rowgroups and columngroups. A rowgroup (columngroup) encompasses several rows (columns). The grouping of rows and columns into g_k rowgroups and g_s columngroups, respectively, is used to achieve ED across the field. We will henceforth refer to the grouping of rows or columns as t -Latinization (John & Williams, 2002). The following description is for columngroups, but the same types of structure can be imposed for the rows, and we do so in the model-based approach described later.

To accommodate arbitrary numbers of columns, we form columngroups from entire columns, meaning that each columngroup will be rectangular in shape. When the number of columns is not divisible by the number of columngroups, this strategy will lead to rectangular columngroups with a varying number of columns. For example, with $s = 11$ columns and $g_s = 3$ columngroups, we can have two columngroups consisting of four columns and one columngroup consisting of three columns (Figure 2). If columngroups need to be unequal in size because the number of columns, s , is not divisible by the number of columngroups, g_s , then we can find two columngroup sizes differing by only one column. We found empirically that placing the smaller columngroups in the interior of the layout provides satisfactory results, so we use that strategy.

3.2 | Evaluation criteria

The following evaluation criteria will be used to compare competing designs:

- The average efficiency factor, $\bar{f}_A^{RC}$, which can be computed as the average pairwise variance of a completely randomized design with the same number of replications per treatment as the current design, divided by the average pairwise variance of the current design ($\bar{f}_A$ in Williams & Piepho, 2015), assuming a model with fixed row and column effects. This is referred to as baseline efficiency (Piepho et al., 2016).
- ED is evaluated using the Euclidean distances (again in terms of row and column numbers) among plots allocated to the same treatment. A measure of ED, denoted as “degree of spreading” by Tedin (1931), is the arithmetic mean length of the lines of a minimum spanning tree connecting all replications of the i -th treatment, denoted as MST_i . The minimum spanning trees were computed using the SAS macro MST (Moser, 1992). We propose to report the smallest MST_i for each number of treatment replications occurring for a design.
- A measure of NB is n_h = number of treatment pairs with h adjacencies ($h = 0, 1, 2, \dots$) in rows. We only report n_h starting from the smallest up to the largest h with $n_h > 0$.

The ED and NB criteria target the two desirable properties we would like to achieve. At the same time, it is important to achieve a good value of the average efficiency factor. The main challenge is to strike the balance between these three criteria, and it is clear that for the improved ED and NB there is a premium to be paid in terms of row-column efficiency compared to the classical row-column design. The next two sections describe our two proposed design strategies. The first optimizes efficiency with respect to a linear mixed model with variance-covariance structure designed to achieve good ED and NB properties, while also obtaining good row-column efficiency. The second directly optimizes ED and NB, while simultaneously seeking to minimize the loss in row-column efficiency relative to a fully randomized row-column design.

3.3 | Design generation using a linear-mixed model

For generating a design, we assume a linear-mixed model of the form

$$\mathbf{y} = \mathbf{X}\boldsymbol{\tau} + \mathbf{f},$$

where $\mathbf{y}$ is the response vector, $\mathbf{X}$ is the treatment design matrix, $\boldsymbol{\tau}$ is the vector of treatment means, and $\mathbf{f}$ is a residual disturbance term comprising all design effects and the residual plot error. Thus, all design effects, such as those for rows and columns, are taken as random. The design search seeks to optimize the treatment information matrix $\mathbf{X}^T \mathbf{V}^{-1} \mathbf{X}$, where $\mathbf{V}$ is the variance-covariance matrix of the residual $\mathbf{f}$. We implement the design search using the SAS procedure OPTTEX, which is part of the QC module and optimizes D-efficiency under the assumed model. The variance-covariance matrix $\mathbf{V}$ for given values of the variance parameters is generated using the MIXED procedure (Piepho, 2015). The variance-covariance structure is designed to achieve good ED and NB properties. At the same time, as the variance-covariance structure also comprises random row and column effects, the design search is expected to also achieve good baseline efficiency $\bar{f}_A^{RC}$.

The residual $\mathbf{f}$ is decomposed into design effects and residual plot error using the model

$$\mathbf{f} = \mathbf{Z}_R \mathbf{u}_R + \mathbf{Z}_C \mathbf{u}_C + \mathbf{Z}_{RG} \mathbf{u}_{RG} + \mathbf{Z}_{CG} \mathbf{u}_{CG} + \mathbf{e},$$

where $\mathbf{u}_R$ is the vector of row effects with variance $\mathbf{I}\boldsymbol{\gamma}_R$, $\mathbf{Z}_R$ is the corresponding design matrix, $\mathbf{u}_C$ is the vector of column effects with variance $\mathbf{I}\boldsymbol{\gamma}_C$, $\mathbf{Z}_C$ is the corresponding design matrix, $\mathbf{u}_{RG}$ is the vector of rowgroup effects with variance $\mathbf{I}\boldsymbol{\gamma}_{RG}$, $\mathbf{Z}_{RG}$ is the corresponding design matrix, $\mathbf{u}_{CG}$ is the vector of columngroup effects with variance $\mathbf{I}\boldsymbol{\gamma}_{CG}$, $\mathbf{Z}_{CG}$ is the corresponding design matrix, and $\mathbf{e}$ is the vector of plot errors with variance-covariance matrix $\mathbf{R}$. Thus, the variance-covariance matrix $\mathbf{V}$ takes the form

$$\mathbf{V} = \mathbf{Z}_R \mathbf{Z}_R^T \boldsymbol{\gamma}_R + \mathbf{Z}_C \mathbf{Z}_C^T \boldsymbol{\gamma}_C + \mathbf{Z}_{RG} \mathbf{Z}_{RG}^T \boldsymbol{\gamma}_{RG} + \mathbf{Z}_{CG} \mathbf{Z}_{CG}^T \boldsymbol{\gamma}_{CG} + \mathbf{R}.$$

For design generation, we consider assignments of different values to the variances for row and column block factors. The larger the variance of a block effect, the larger its influence on the design. Note that as the variance approaches infinity, the effect behaves as a fixed effect. For the variance of plot errors ($\mathbf{R}$) we either assume a baseline structure with $\mathbf{R} = \mathbf{I}\boldsymbol{\gamma}_E$, or one of two possible spatial variance-covariance structures. The first is intended to optimize ED, whereas the other is designed to optimize NB. The two spatial variance-covariance structures are as follows:

1. A two-dimensional isotropic power model for spatial covariance assuming that plots are correlated across the whole trial area, using row and column numbers as the coordinates. This model is aimed at ED across the experimental field. The spatial covariance among two plots 1 and 2 is $\gamma_{ED}\rho_{ED}^{d_{12}}$, where γ_{ED} is the plot variance, ρ_{ED} is the auto-correlation, and d_{12} is the Euclidean distance between the two plots in terms of row and column numbers, that is, $d_{12} = \sqrt{(i_1 - i_2)^2 + (j_1 - j_2)^2}$, where the first plot is in row i_1 and column j_1 and the second plot is in row i_2 and column j_2 .
2. A one-dimensional model assuming that adjacent plots are correlated along the same row only. This model is primarily aimed at NB within rows. The variance is γ_{NB} , whereas the covariance is equal to zero for nonneighbors and $\gamma_{NB}\rho_{NB}$ for nearest neighbors (Kiefer & Wynn, 1981). This is the same as a banded Toeplitz variance-covariance structure with nonzero covariance only on the first off-diagonal (Wolfinger, 1993). This design option ignores the treatments that are present on the plots bordering the trial from the outside at both ends of the rows.

As an alternative to option (2), one may use a modified version of the banded Toeplitz model, again assuming that plots are correlated along the same row only. This model [option (2b)] is aimed at NB within rows when we have a choice as to which treatment is placed on the outside border plots present at both ends of the rows. Note that generally these border plots do not themselves enter the statistical analysis but treatments allocated to them are chosen deliberately so that plots at both row ends have suitable outside neighbors on both sides (Rees, 1967). This model is the same as that described under (2), except that the covariance is equal to $\gamma_{NB}\rho_{NB}$ for the two plots at the two ends of the same row. The rationale for this approach is that the design problem is equivalent to that for circular blocks (Rees, 1967; see Appendix A). While the design problem itself is well known and many solutions have been proposed for special cases (e.g., Azais et al., 1993), the covariance-based numerical design search approach suggested here does not seem to have been used before for this type of spatial variance-covariance structure. An example is considered in the Appendix A.

As we pointed out in the introduction, complete NB usually requires a larger number of replications than is feasible or desirable. In most practical settings one can only hope for partial NB, that is, a design where the numbers of adjacencies per treatment pair take on only two different values differing by unity. For details on when complete NB is possible under options (2) and (2b) see the Appendix A. In the main body of the paper, we will exclusively use option (2).

Without loss of generality we here set $\gamma_{ED} = \gamma_{NB} = 1$. We have found from extensive empirical experience that $\rho_{ED} = \rho_{NB} = 0.1$ works well with all three options. This value of the correlation has also been used with a one-dimensional power model in Example 14.9 of the online documentation for the OPTEx procedure. We have also found the combination of structures (1) for ED and (2) for NB to be useful. When combining two residual structures, we simply add the two structures, meaning that the total error variance equals $\gamma_{ED} + \gamma_{NB} = 2$.

3.4 | Direct optimization of ED and NB

We here describe an interchange algorithm that has been implemented to directly improve the ED and NB of a given design. The algorithm starts with an efficient row-column design and subsequently seeks to improve nine spatial scores that count the number of occurrences of undesirable treatment allocations to plots neighboring a target plot. Six of these configurations correspond to self-adjacencies illustrated in Figure 3 and described in Table 1. The nine spatial scores (S1–S9), listed in the order of priority they are improved by the algorithm, are defined in Table 2. The score S1 addresses NB, whereas the other eight scores target the ED of a design. The algorithm also considers two further criteria targeting ED, that is, the span along columns and rows. The span is defined as the maximum number of columns or rows two replicates of a treatment are apart. For example, if a treatment appears in columns 3, 5, 6, and 8, the span is six columns. An initial design is generated considering scores S1 to S9. Subsequently, the algorithm seeks to improve the minimum row and column spans, as well as baseline efficiency factor $\bar{f}_A^{RC}$, making sure that S1 and S2 are not reduced. In fact, it may be regarded as an advantage of this approach that the baseline efficiency is directly targeted. At this later stage, criteria S3 to S9 are not tracked because our preliminary experience has shown that this would not leave enough room to improve the minimum row and column span. Thus, the main emphasis of design optimization by the direct method is on S1, S2, row minimum span, column minimum span, and $\bar{f}_A^{RC}$.

We require that $v \geq s \geq k$ and a binary starting design. A design is binary if both rows and columns are binary. No interchanges are accepted that make the design nonbinary. Two methods are used, namely the divisible and filter methods. Which method is used depends on the values of v , k , and s . Let the integer n be defined to be such that $nk \leq v$ and $(n+1)k > v$. Then if $nk = v$ and n divides s , the divisible method is used. Otherwise if $n = 1$ or $n = 2$ the filter method is used. In all other cases the divisible method is used. For each starting design the algorithm runs as follows:

		3		3		
	4	2		2	4	
6	5	1		1	5	6
			×			
6	5	1		1	5	6
	4	2		2	4	
		3		3		

FIGURE 3 Illustration of six different types of self-adjacencies (SA1-SA6) relative to the target plot (×). See text for detailed description. The direct optimization approach seeks to avoid these six types of self-adjacencies. The figure does not highlight self-adjacencies in the same row or column because the design is assumed to be binary

TABLE 1 Description of six types of self-adjacency (SA1 to SA6) (also see Figure 3)

Type of self-adjacency	Description
SA1	Single diagonal
SA2	Knight move in column direction (two plots in column direction, one plot in row direction)
SA3	Stretched knight move in column direction (three plots in column direction, one plot in row direction)
SA4	Double diagonal
SA5	Knight move in row direction (two plots in row direction, one plot in column direction)
SA6	Stretched knight move in row direction (three plots in row direction, one plot in column direction)

TABLE 2 Description of nine spatial scores (S1–S9) optimized by our direct method

Spatial score	Description
S1	Number of SA1 self-adjacencies
S2	Number of row-neighbor occurrences; if a treatment pair has n row-neighbor occurrences, its contribution to the score is $n(n-1)/2$
S3	Number of SA2 self-adjacencies
S4	Number of times a treatment is repeated in an adjacent column
S5	Number of SA3 self-adjacencies
S6	Number of SA4 self-adjacencies
S7	Number of SA5 self-adjacencies
S8	Number of SA6 self-adjacencies
S9	Number of times a treatment is repeated in columns two apart

1. If the divisible method is used, then the columns are divided into columngroups in such a way that the number of plots in any columngroup is less than or equal to v . Interchanges are then made with the aim of minimizing the number of duplicated treatments in a columngroup. If the number of plots in each columngroup is equal to v then it is very likely that the resulting design will be resolvable, although there is no guarantee that this will be the case.
2. Interchanges are then carried out to try to optimize the value of a given spatial criterion (Table 2), without degrading any earlier criteria. The priority order is S1 through to S9. If the divisible method is chosen, interchanges are only carried out within columngroups, while for the filter method there are no restrictions on interchanges.

3. If the divisible method is used, interchanges—within rows and columngroups—then take place to try to improve treatment column spans without affecting the values of the S1 and S2 criteria. The same procedure is then carried out within columns in an attempt to improve treatment row spans. In other words these procedures tweak columns and rows.
4. If the filter method is used, then the columns are divided into columngroups such that the number of columns in the columngroup does not differ by more than one. For equally replicated designs where the number of replicates r divides s then each columngroup will contain s/r columns. Interchanges are then made with the aim of minimizing the number of duplicated treatments in a columngroup, without degrading the values of the S1 to S4 criteria.

Next the program determines whether to accept or reject the design found. The steps involved are as follows:

1. The program subsequently determines whether to accept or reject the design found. A design is accepted only if its scores for S1, S2, minimum row span and minimum column span are not worse than the previous best value found for these criteria.

The starting design is now randomized to produce a new starting design and the program returns to step 1 above. This whole procedure is repeated 50 times. When this stage is completed the starting designs continue to be randomized with the program returning to step 1 each time. Now, however, there is an additional step involved in determining whether to accept or reject the spatial design. After step 5 the average efficiency factor $\bar{f}_A^{RC}$ of the row-column design is calculated leading to the following:

1. If $\bar{f}_A^{RC}$ is less than the previous best $\bar{f}_A^{RC}$ value found the design is rejected.
2. Otherwise, the design is accepted and stored and the values of the criteria in steps 5, as well $\bar{f}_A^{RC}$, replace the best values previously found.

The algorithm prints summary lines during the updating process. Improvements usually occur rapidly at the start of the process and then stabilise, possibly with infrequent and minor updates. For smaller arrays this stable position is achieved in seconds; larger arrays should be run for several minutes. The user can set an initial seed to allow for reproducibility. The algorithm continues until the user intervenes by pressing any button on the keyboard upon which the final design is output, along with a count number for the iteration step at which the user intervened, thus ensuring full reproducibility.

We note that if one starts with a resolvable design, then the algorithm will always keep that resolvability. On the other hand, starting with a nonresolvable design may or may not result in a resolvable design at the end (i.e., in cases where the parameters allow resolvability).

3.5 | Analysis

Both design strategies involve random interchanges and therefore directly generate a randomized design. Due to the search strategy and criteria used, however, the randomization set is clearly restricted and a purely randomization-derived linear model is not available. Hence, the choice of analysis model is not unequivocal. We propose to start analysis using a linear model with effects for treatments, rows, columns, and any super-imposed blocks. Although our design-generation methods impose additional restrictions on randomization compared with classical row-column designs, empirical evidence based on the analysis of uniformity data suggests that analysis based on a row-column linear model will provide approximately valid inferences (Williams & Piepho, 2018). It can then be investigated whether addition of spatial covariance structures (Gilmour, Cullis, & Verbyla, 1997) or neighbor effects (Bailey & Druilhet, 2004) improve the fit using likelihood-based criteria (Piepho & Williams, 2010). A more conservative approach is to not perform any model selection and stick with the basic row-column model.

4 | EXAMPLES

4.1 | Application to Example 1

We first apply our model-based strategy to see if the classical design in Figure 1 can be improved upon in terms of ED and NB. We form five rowgroups ($g_k = 5$) and five columngroups ($g_s = 5$). Values of the variance component were chosen by trial-and-error. The design in Figure 4 is improved compared to that in Figure 1 in terms of ED, but the improvement comes at the cost of a slightly deteriorated baseline efficiency. NB is improved as well, leaving only three treatment pairs with two adjacencies in rows, but there still seems to be room for further improvement toward partial NB. If we focus on adjacencies within rows using option (2), there are $(s - 1)$ adjacencies in a row and thus $k(s - 1) = 90$ adjacencies overall. With $v(v - 1)/2 = 300$ treatment

9	2	8	6	15	1	17	13	5	12	7
1	5	7	17	14	12	15	8	6	2	21
4	23	25	20	11	22	3	10	18	24	16
22	7	18	4	2	25	6	23	11	1	10
13	14	24	12	20	3	16	19	8	21	9
16	8	23	9	6	13	12	18	20	22	3
10	24	5	14	4	19	11	21	15	25	17
21	25	10	11	9	17	22	5	19	14	24
3	18	2	15	7	16	1	20	23	13	4

FIGURE 4 Two-dimensional layout for $(v, k, s, g_k, g_s) = (25, 9, 11, 5, 5)$ generated using model-based approach with $(\gamma_R, \gamma_C, \gamma_{RG}, \gamma_{CG}, \gamma_{ED}, \rho_{ED}) = (10^{-2}, 10^{-2}, 1, 1, 1, 10^{-1})$ [option (1) with grouping of rows and columns]. $\bar{f}_A^{RC} = 0.824$; $(n_0, n_1, n_2) = (213, 84, 3)$; $MST_{19} = 3.00$ (treatment 19 is the one with only three replications), $MST_{20} = 3.17$ (smallest value for treatments with four replications)

25	11	18	16	19	20	12	1	10	7	2
12	14	5	10	18	13	2	6	22	19	17
7	15	23	21	16	8	20	24	11	9	4
21	5	1	22	14	2	17	11	8	25	23
6	18	4	19	24	9	10	15	12	3	13
17	4	14	20	3	7	13	16	6	21	1
8	19	25	24	23	12	9	22	15	18	5
9	1	15	2	21	17	8	4	16	14	25
13	23	11	3	10	22	7	5	24	6	20

FIGURE 5 Two-dimensional layout for $(v, k, s, g_k, g_s) = (25, 9, 11, 5, 5)$ generated using model-based approach with $(\gamma_R, \gamma_C, \gamma_{RG}, \gamma_{CG}, \gamma_{NB}, \rho_{NB}) = (10^{-2}, 10^{-2}, 1, 1, 1, 10^{-1})$ [option (2) with grouping of rows and columns]. $\bar{f}_A^{RC} = 0.823$; $(n_0, n_1) = (210, 90)$; $MST_3 = 4.13$ (treatment 3 is the one with only three replications), $MST_{16} = 2.90$ (smallest among treatments with four replications)

pairs it seems possible that designs must exist with a maximum of one adjacency per treatment pair. Thus, we replace the power law for the residual extending across the whole trial [option (1)] with the Toeplitz model for the within-row correlation [option (2)]. The resulting design, shown in Figure 5, indeed has the desirable property of partial NB, with the number of adjacencies within a row equalling either zero or unity. At the same time, ED in terms of MST_i has diminished somewhat, exemplifying the trade-off between ED and NB. Also, there are now two diagonal self-adjacencies, as opposed to the design in Figure 4, which

2	20	15	22	7	4	12	19	11	17	23
9	16	13	14	24	18	6	25	3	5	8
25	24	6	21	19	13	5	10	16	15	18
19	5	7	15	10	24	1	22	8	21	13
14	9	4	20	6	23	11	12	17	2	3
8	23	21	17	4	19	22	6	15	14	7
20	11	10	3	18	9	25	1	2	16	12
12	18	22	16	23	10	8	4	21	25	20
3	1	17	9	13	7	2	5	14	11	24

FIGURE 6 Two-dimensional layout for $(v, k, s, g_k, g_s) = (25, 9, 11, 5, 5)$ generated using model-based approach with $(\gamma_R, \gamma_C, \gamma_{RG}, \gamma_{CG}, \gamma_{ED}, \gamma_{NB}, \rho_{ED}, \rho_{NB}) = (10^{-2}, 10^{-2}, 1, 1, 1, 1, 10^{-1}, 10^{-1})$ [options (1) and (2) combined, with grouping of rows and columns]. $\bar{f}_A^{RC} = 0.817$; $(n_0, n_1) = (210, 90)$; $MST_1 = 4.74$ (treatment 1 is the one with only three replications). $MST_6 = 3.20$ (smallest among treatments with four replications)

has no such adjacency. For the nearest neighbor distances, however, there is a slight improvement for the design in Figure 5 compared to that in Figure 4 in terms of n_h . If we combine options (1) and (2) (Figure 6), a design with partial NB is obtained with no treatment pair appearing next to each other in the same row more than once, and ED almost as good as that in Figure 4. This time, there are no diagonal self-adjacencies. Thus, the combination of options (1) and (2) provides a good compromise, though there is indeed a slight premium to pay in terms of the average efficiency factor as foreshadowed in Section 3.3.

In our experience, the isotropic power model [option (1)] does a good job of achieving ED. For comparison, we also considered an anisotropic model (Figure S1 in Supporting Information) where the spatial covariance among two plots is $\gamma_E \rho_R^{|j_1 - j_2|} \rho_C^{|i_1 - i_2|}$, where ρ_R and ρ_C are the autocorrelations along the rows and columns, respectively. The anisotropic power model with distances defined based on row and column number is equivalent to the separable $AR1 \otimes AR1$ model, which is quite popular in both the design and analysis of field trials (Gilmour et al., 1997). This model leads to an inferior ED and NB. Remarkably, the design in Figure S1 has 14 diagonal self-adjacencies, whereas with the isotropic power model there is no such adjacency (Figure 4). Eccleston and Chan (1988) also used the $AR1 \otimes AR1$ model and reported presence of diagonal self-adjacencies. The reason for this behavior of the anisotropic model is the lower correlation of diagonally adjacent plots. For example, when the correlation equals 0.1, the correlation among diagonally adjacent plots is 0.01 for the anisotropic model and 0.04 for the isotropic model, whereas the correlations among adjacent plots in the same row or column equal 0.1 for both models. Due to the much lower correlation with the anisotropic model, there is a lower penalty for diagonal self-adjacencies, making these more likely to occur in a generated design. With higher values for the autocorrelation the problem with diagonal self-adjacencies persisted though it was not as pronounced (results not shown).

Next, we applied the direct design optimization algorithm described in Section 3.4. The resulting design is shown in Figure 7. It compares quite favorably with the model-based design in Figure 6 in terms of both ED and NB and the value of the MST criterion is best. Also, the loss in the average efficiency factor $\bar{f}_A^{RC}$ compared to the classical design in Figure 1 is smallest. Both designs (Figures 6 and 7) have $(n_0, n_1) = (210, 90)$, which is optimal in terms of neighbor balance.

4.2 | A second example

Example 2: Consider a design with $s = v = 8$ and $k = 4$. The classical row-column design in Figure S2 has baseline efficiency $\bar{f}_A^{RC} = 0.850$, but does not fare well in terms of NB in rows or ED. For example, the neighbor pair (8,2) occurs four times, and treatment 3 has its four replications on a single diagonal. In this case, complete NB is possible (see Appendix A) and our

14	5	9	17	7	16	6	2	3	25	15
16	1	18	13	8	12	22	9	10	20	6
19	24	22	10	25	14	18	15	13	5	21
20	8	21	6	9	4	7	23	19	2	11
25	17	16	12	19	21	10	8	22	18	24
11	15	7	5	1	2	24	12	17	4	14
12	23	4	20	11	3	5	25	16	9	8
13	10	2	14	23	17	20	21	1	7	3
18	6	3	15	24	13	19	4	11	23	22

FIGURE 7 Two-dimensional layout for $(v, k, s, g_s) = (25, 9, 11, 4)$ generated using the filter method with our direct approach. $\bar{f}_A^{RC} = 0.841$; $(n_0, n_1) = (210, 90)$; $MST_1 = 4.74$ (treatment 1 is the one with only three replications), $MST_{12} = 3.78$ (smallest among treatments with four replications)

2	6	5	8	4	7	1	3
7	8	1	2	3	6	4	5
5	3	7	4	8	1	2	6
4	1	6	3	2	5	7	8

FIGURE 8 Two-dimensional layout for $(v, k, s, g_s) = (8, 4, 8, 4)$ generated using model-based approach with $(\gamma_R, \gamma_C, \gamma_{CG}, \gamma_{ED}, \rho_{ED}) = (1, 10^{-3}, 10^{-2}, 1, 10^{-1})$ [option (1) with grouping of columns]. $\bar{f}_A^{RC} = 0.801$; $(n_0, n_1, n_2) = (8, 12, 8)$; $MST_1 = 2.54$

model-based approach finds such a design, with each pair of treatments occurring exactly once in a row (Figure S3). Note that we had to reduce the column variance to $\gamma_C = 10^{-3}$ in order to push the design to complete NB. The baseline efficiency is reduced compared to a classical row-column design. We also note that the design still has a considerable number of diagonal self-adjacencies. These can be avoided by focusing on ED using option (1) and grouping columns into four groups of two columns each, but at the cost of giving up complete NB (Figure 8). The resulting design is a resolvable row-column design with groups of two columns forming a complete replicate. The direct approach finds the design in Figure 9, which shows good ND and ED properties comparable to that in Figure 8. The design also has complete replicates formed by pairs of columns. Its baseline efficiency is better than that of the design in Figure 8, and it has six treatment pairs appearing as neighbors twice ($n_2 = 6$), as compared to $n_2 = 8$ for Figure 8.

4.3 | A third example

Example 3: In addition to the rowgroups and columngroups as employed in our approach, further superblocks can be introduced that span several subrows and subcolumns. A convenient way to generate these superblocks is to define a second set of rowgroups and columngroups and form superblocks by their intersection. Introducing this third blocking factor leads to designs akin to *gerechte designs* as proposed by Behrens (1956a,b; also see Bailey, Kunert, & Martin, 1990, 1991) to allow for a “fair distribution” (*gerechte Verteilung*) of treatment replication (Schuster & Lochow, 1992). Figure 10 shows such a design obtained

3	6	2	8	1	4	5	7
1	4	5	3	2	7	8	6
5	7	6	1	8	3	4	2
2	8	4	7	6	5	1	3

FIGURE 9 Two-dimensional layout for $(v, k, s, g_s) = (8, 4, 8, 4)$ generated using the divisible method with our direct approach. $\bar{f}_A^{RC} = 0.840$; $(n_0, n_1, n_2) = (6, 16, 6)$; $MST_7 = 2.43$

6	8	15	4	13	10	12	3	7	9
11	2	14	1	5	15	13	6	4	8
3	9	12	7	10	11	1	2	14	5
1	10	8	14	6	9	7	13	11	15
5	4	2	12	3	14	8	1	6	10
7	13	9	15	11	5	4	12	2	3

FIGURE 10 Two-dimensional layout for $(v, k, s, g_k, g_s) = (15, 6, 10, 4, 4)$ generated using model-based approach with $(\gamma_R, \gamma_C, \gamma_{RG}, \gamma_{CG}, \gamma_B, \gamma_{ED}, \gamma_{NB}, \rho_{ED}, \rho_{NB}) = (10^{-1}, 10^{-1}, 10^4, 10^4, 10^4, 1, 1, 10^{-1}, 10^{-1})$ [option (1) and (2) combined]. $\bar{f}_A^{RC} = 0.831$; $(n_0, n_1, n_2) = (59, 38, 8)$; $MST_{14} = 2.69$

with our model-based approach, assuming a variance $\gamma_B = 10^4$ for four *rectangular* super-imposed blocks, delineated by two perpendicular red lines. Moreover, there are four *crooked* rowgroups (two of which are highlighted in gray) and four *crooked* columngroups (delineated by the thick black lines and the central thick red line). The formation of such crooked blocks is described in Appendix B. The separation of treatment replications is better than that for a classical row-column design shown in Figure S4, where there are several diagonal self-adjacencies and treatment 7 shows a rather extreme form of clumping. By comparison, the design in Figure 10 has no diagonal self-adjacency. The design in Figure 11 was obtained using our direct approach, which does not involve super-imposition of a third super-block structure for *gerechte designs*, but only uses $g_s = 4$ columngroups. Nevertheless, it does very well in terms of ED and NB.

5 | DISCUSSION

The ad hoc modifications of RCBD that are commonly encountered in practice often target ED or NB or a combination of both, though not in a systematic and objective way. The advantage of our proposed procedures is that they use design criteria to optimize ED and NB by using numerical searches. As such, they are clearly preferred to ad hoc modifications of RCBD.

At the time when a trial needs to be designed there is usually little if any information about the sources of spatial variation to be expected and their likely magnitudes. Thus, a design strategy is desirable that can cater for diverse sources of error, rather than focussing on one specific environmental factor. The typically occurring sources of error fall into the following three categories:

- Large-scale spatial trends, usually continuous and irregular
- Effects associated with rows or columns, usually caused by tractor-driven equipment for farming operations (tillage, sowing, disease control, fertilization, harvesting)
- Neighbor effects between treatments on adjacent plots (mostly within rows)

5	2	3	12	8	7	4	11	6	1
14	10	11	7	9	1	13	15	12	2
9	4	8	15	10	6	3	5	13	11
7	5	6	13	3	8	2	14	4	15
2	1	14	5	4	12	10	9	8	6
10	13	12	11	14	9	15	1	7	3

FIGURE 11 Two-dimensional layout for $(v, k, s, g_s) = (15, 6, 10, 4)$ generated using the filter method with our direct approach. $\bar{f}_A^{RC} = 0.832$; $(n_0, n_1) = (51, 54)$; $MST_{13} = 2.89$

These three sources of error are tackled by the following three components of our search strategies:

- (a) Latinization/*t*-Latinization and imposition of a spatial error model ensure that treatment replications are well distributed across the whole experimental area (ED), avoiding clustering of replication in a restricted area of the trial
- (b) Row and column blocking accounts for effects introduced by management operations
- (c) Imposition of spatial covariance within rows improves NB and ED of the design (model-based approach)/interchanges of treatments to optimize a suite of criteria (Tables 1 and 2) (direct approach)

Our experience with the proposed procedures is that the three major objectives, that is, ED, row-column blocking and NB within rows, can be achieved simultaneously. This, in turn, provides the chances that errors occurring during the course of a trial can be well captured by the design and corrected for during analysis using a suitable mixed model. We hope that these procedures will therefore make row-column designs more appealing to research scientists and thus will foster the more widespread use of this class of designs, especially in settings where neighbor effects or irregular patterns of trend or even loss of clusters of plots are expected. The model-based method described in this paper has been implemented in a SAS macro that is, available from the authors on request. The macro has been integrated into the analysis system PIAFStat in operation for routine analysis of regional postregistration and agronomy trials conducted by the federal states in Germany (Schmidtke & Zink, 2007). The direct approach is available as a stand-alone executable file that will be integrated into a future release of CycDesignN (VSN International Inc.).

The starting point for our proposed approach to design construction is a classical row-column design. This blocking structure accounts for major sources of variation to be expected from trial operations conducted along rows and columns. The design is not required to be resolvable, though resolvability, which is generally desirable, can be achieved when the dimensions of the blocking units of the trial permit. ED may be targeted by grouping rows and columns into larger units, for which additional random effects are fitted. In case of a resolvable design, grouping can be done in a way leading to a *t*-Latinized layout (John & Williams, 2002). The novelty here is that grouping/*t*-Latinization can also be performed in case of a nonresolvable design.

With our model-based approach, an isotropic power model with spatial covariance between plots extending across the whole field usually improves ED. NB within rows may be addressed by changing the plot error model to a Toeplitz structure with covariance extending across plots in the same row (Kiefer & Wynn, 1981). There are two variants to the model, depending on whether or not the treatment planted on the border plots are to be taken into account in evaluating NB [options (2) and (2b)]. The choice of plot error covariance model as well as the choice of values assigned to the variance components determines whether the primary thrust of the design search is in the direction of ED or of NB. It is admitted here that our choice of variance component values is ad hoc in nature, and we have no specific method to offer other than what is best denoted as trial-and-error. Obviously, the relative importance given to a random model term depends on the magnitude of the assigned variance, but how exactly a particular choice of values plays out must be investigated in each case.

Neighbor-balanced designs make sure that each treatment will have each other treatment as a neighbor the same number of times (see Appendix A). If one treatment has a strong neighbor effect then all other treatments are equally affected because each other treatment appears as a neighbor exactly once. This particular type of perfect balance may be desirable where it is feasible. At the same time it must be acknowledged that bias resulting from a strong neighbor effect of one treatment compared to the other treatments cannot be annihilated by NB; the bias of the one treatment with a strong neighbor effect compared to other

treatments can be alleviated, however, by explicitly modeling neighbor effects (Bailey & Druilhet, 2004), but this option is not considered here. In the trials we are focussing on here, partial NB is mainly an insurance policy against adverse effects resulting from complete failure of a treatment, for example, due to frost damage. In such cases, it may be necessary to drop all neighboring plots in the same row from the analysis. Partial NB then ensures that the plots that need to be dropped from analysis are shared evenly among the other treatments. The main objective therefore is to avoid a large number of occurrences of the same treatment pairs as row neighbors. With this objective in mind, it makes little practical difference whether option (2) or (2b) is used with the model-based approach to achieve some degree of NB. Also, more often than not dimensions of the trial are such that partial NB is the only feasible objective, in which case the choice between (2) and (2b) is of secondary practical relevance. In our four examples we have exclusively focused on option (2).

Most publications on NB consider designs with one blocking factor, but some explicitly look into NB in row-column designs (see Varghese et al., 2014, and references therein). Chan and Eccleston (1998) consider an extension to NB along rows and columns. They refer to these designs as two-dimensional “nearest neighbor balanced designs.” They consider complete and partial NB, making a distinction between designs with and without border plots taken into account. Our approach can be extended to cover nearest neighbors in two dimensions by adding a covariance among plots in the same column, but we have not pursued this here because column neighborhoods are not a problem with the types of trials with cereals and other field crops we are focussing on, which are characterized by plots that are longish down the columns and where rows are separated by tractor driveways.

Compared to our model-based approach, the direct approach makes no use of a specific model assumption for design generation, but baseline efficiency of the row-column design is monitored throughout the search and when the divisible method is used, columngroups and rowgroups are maintained. The method successfully achieves ED and NB by directly targeting a set of criteria outlined in Tables 1 and 2. For designs with a large number of treatments, these criteria target the immediate vicinity of a plot in terms of ED and NB, whereas more distant plots are not considered even if located in the same replicate, rowgroup, or columngroup. One major advantage of the direct approach is that assignment of suitable variance values as in the model-based approach is obviated. A further advantage is that baseline efficiency is directly optimized (within the constraints imposed by the other criteria). This can be seen in the examples, where the designs generated by the direct approach have better baseline efficiency than those generated by the model-based approach.

Both approaches proposed here yield designs that are improved compared to classical row-column designs in terms of ED and NB. The model-based approach has the advantage of flexibility, allowing us to direct the focus on either ED or NB or on both simultaneously, by suitable choice of the variance components and the spatial variance-covariance structure for the residual. The direct approach has the virtue of simplicity as no fine-tuning is necessary because it does not make explicit use of a mixed model, though it does account for the row-column block structure and superimposed rowgroups and columngroups. We would refrain from making a strong recommendation in favor of one or other of the two approaches because both have their merits and deserve being considered for design generation. It should be reiterated that with both approaches there is a premium to be paid, in terms of a slight loss in baseline efficiency $\bar{f}_A^{RC}$, for the improved ED and NB. With larger designs, however, the loss is usually small compared to unconstrained randomization, so the improved ED and NB can be had essentially for free.

The strategies advocated here restrict the randomization set compared to complete randomization as used with classical row-column designs. This restriction is likely to cause an upward bias in the error variance estimate when a classical row-column model is used for analysis. Tedin (1931) investigated this for Latin square designs. Using uniformity data he showed empirically that Latin squares with a knight move layout for treatment replications have larger expected error mean square than under full randomization according to a Latin square. Knight move designs are of the form as targeted by our strategy in that the spread of treatment replications is maximized. Tedin's empirical evidence is for 5×5 designs and the bias is modest. It seems reasonable to expect that bias is smaller for larger designs. Also, unless the design is very small, the randomization set under restricted randomization as imposed by our design strategy will be large enough to allow a meaningful statistical analysis. If a classical estimate of the error variance is used, the true error variance will tend to be overestimated, so inference can be conservative. Then by adding effects for groups of rows and columns and a spatial covariance component, the true error variance may be hoped to be better approximated in the analysis.

Also, one may consider the definition of the randomization set. This is usually defined to have the same average efficiency factor (i.e., exactly the same design quality) for each design in the set. With search methods, however, there can often be large numbers of designs near the optimum that only differ in average efficiency factor in the fourth (or more) significant figure. From a practical viewpoint a set of these could be viewed as a reasonable “randomization set” (Williams & Piepho, 2018). In principle, statistical inference could be based on a computer-based generation of the randomization set, but we prefer the linear-model based approach outlined in Section 3.5 for its simplicity and versatility.

van Es, Gomes, Sellmann, and van Es (2007) advocate so-called spatially balanced complete block designs, where the allocation of treatments to plots within blocks follows two objectives: (i) the average within-block distance is about the same

for each treatment pair and (ii) treatments are not allowed to occur at the same position in different blocks (unless the number of replications is larger than the number of treatments). They point out that their objective (i) implies a linear variance model for spatial correlation within blocks, but they do not use this model explicitly. Instead, their objective criterion is based on the variance between treatment pairs of the average within-block distance between treatments. Their algorithm is computationally rather intensive, and they provided tables with designs for up to 15 blocks and 15 treatments. Using our model-based approach, the linear variance model can be specified explicitly in place of the auto-regressive models we used here (Piepho et al., 2016). Also, when rows correspond to complete blocks, imposing a column effect with small variance will ensure that treatments do not occur more than once in the same position of a complete block. Using this approach, we obtain designs similar to those reported in van Es et al. (2007). For example, van Es et al. (2007) report the design (ABCD), (CDAB), (DABC), (BCDA) for $v = 4$ treatments and $k = 4$ complete blocks, where A–D are the treatment labels and brackets embrace complete blocks. Using our model-based approach, we find the design (BCDA), (ADCB), (CABD), (DBAC). For both designs, the average within-block distance is 2.0 for two treatment pairs and 1.5 among the other four pairs. The main difference is that our design is neighbor balanced with two adjacencies for each pair, whereas that of van Es et al. (2007) has three adjacencies for four pairs and no adjacencies at all for two pairs.

ACKNOWLEDGMENTS

We thank Nye John for implementing the direct search algorithms described in Section 3.4.

CONFLICT OF INTEREST

The authors have declared no conflict of interest.

ORCID

Hans-Peter Piepho  <http://orcid.org/0000-0001-7813-2992>

REFERENCES

- Azais, J. M., Bailey, R. A., & Monod, H. (1993). A catalogue of efficient neighbour-designs with border plots. *Biometrics*, 49, 1252–1261.
- Bailey, R. A., & Druihet, P. (2004). Optimality of neighbor-balanced designs for total effects. *The Annals of Statistics*, 32, 1650–1661.
- Bailey, R. A., Kunert, J., & Martin, R. J. (1990). Some comments on gerechte designs. I. Analysis for uncorrelated errors. *Journal of Agronomy and Crop Science*, 165, 121–130.
- Bailey, R. A., Kunert, J., & Martin, R. J. (1991). Some comments on gerechte designs. II. Randomization analysis, and other methods that allow for inter-plot dependence. *Journal of Agronomy and Crop Science*, 166, 101–111.
- Behrens, W. U. (1956a). Feldversuchsanordnungen mit verbessertem Ausgleich der Bodenunterschiede. *Zeitschrift für Landwirtschaftliches Versuchs- und Untersuchungswesen*, 2, 176–173.
- Behrens, W. U. (1956b). Die Eignung verschiedener Feldversuchs-Anordnungen zum Ausgleich der Bodenunterschiede. *Zeitschrift für Acker- und Pflanzenbau*, 101, 243–278.
- Chan, B. S. P., & Eccleston, J. A. (1998). On the construction of complete and partial nearest neighbour balanced designs. *Australasian Journal of Combinatorics*, 18, 39–50.
- Chan, B. S. P., & Eccleston, J. A. (2003). On the construction of nearest neighbour balanced row-column designs. *Australian and New Zealand Journal of Statistics*, 45, 97–106.
- Cheng, C. S. (1983). Construction of optimal balanced incomplete block designs for correlated observations. *The Annals of Statistics*, 11, 240–246.
- Eccleston, J. A., & Chan, B. S. P. (1998). *Design algorithms for correlated data* (pp. 41–52). In R. Payne & P. Green (Eds.), *COMPSTAT 1998: Proceedings in Computational Statistics*. New York: Heidelberg. Physica-Verlag.
- Federer, W. T., Nair, R. C., & Raghavarao, D. (1975). Some augmented row-column designs. *Biometrics*, 31, 361–375.
- Federer, W. T., & Basford, K. (1991). Competing effects designs and models for two-dimensional arrangements. *Biometrics*, 47, 1461–1472.
- Gilmour, A. R., Cullis, B. R., & Verbyla, A. P. (1997). Accounting for natural and extraneous variation in the analysis of field experiments. *Journal of Agricultural, Biological and Environmental Statistics*, 2, 269–293.
- Jaggi, S., Varghese, C., Varghese, E., & Sharma, A. (2015). Web generation of experimental designs balanced for indirect effects of treatments (WEB-DBIE). *Computers and Electronics in Agriculture*, 111, 52–68.
- John, J. A., & Williams, E. R. (1995). *Cyclic and computer generated designs* (2nd ed.). London, UK: Chapman and Hall.
- John, J. A., & Williams, E. R. (2002). t-latinized designs. *Australian and New Zealand Journal of Statistics*, 40, 111–118.

- Kiefer, J., & Wynn, H. P. (1981). Optimum balanced block and Latin square designs for correlated observations. *The Annals of Statistics*, 9, 737–757.
- Lawless, J. F. (1971). A note on certain types of BIBD's balanced for residual effects. *The Annals of Mathematical Statistics*, 42, 1439–1441.
- Moser, E. B. (1992). Ordination and display of multivariate data. In: SAS Institute Inc. (pp. 1345–1355). Proceedings of the Seventeenth Annual SAS Users Group International Conference, Cary, NC: SAS Institute Inc.
- Piepho, H. P. (2015). Generating efficient designs for comparative experiments using the SAS procedure OPTEX. *Communications in Biometry and Crop Science*, 10, 96–114.
- Piepho, H. P., Michel, V., & Williams, E. R. (2015). Beyond Latin squares: A brief tour of row-column designs. *Agronomy Journal*, 107, 2263–2270.
- Piepho, H. P., Michel, V., & Williams, E. R. (2016). Nonresolvable row-column designs with an even distribution of treatment replications. *Journal of Agricultural, Biological and Environmental Statistics*, 21, 227–242.
- Rees, D. H. (1967). Some designs of use in serology. *Biometrics*, 23, 779–791.
- Schmidtke, J., & Zink, G. (2007). *PIAF/PIAFStat (Planning-, information- and evaluation-system for field trials)* (pp. 203208). In H. P. Piepho & H. Bleiholder. (Eds.), *Agricultural field trials—today and tomorrow*. Proceedings of the International Symposium 08–10 October 2007, Stuttgart-Hohenheim, DE: Beuren: Verlag Grauer.
- Schuster, W., & von Lochow, J. (1992). *Anlage und Auswertung von Feldversuchen*. 3. Auflage. Strothe, Frankfurt: Verlag Alfred.
- Smith, A. B., Lim, P., & Cullis, B. R. (2006). The design and analysis of multi-phase plant breeding experiments. *Journal of Agricultural Science*, 144, 393–409.
- Tedin, O. (1931). The influence of systematical plot arrangement upon the estimate of error in field experiments. *Journal of Agricultural Science*, 21, 191–208.
- van Es, H. M., Gomes, C. P., Sellmann, M., & van Es, C. L. (2007). Spatially-balanced complete block designs for field experiments. *Geoderma*, 140, 346–352.
- Varghese, E., Jaggi, S., & Varghese, C. (2014). Neighbor-balanced row-column designs. *Communications in Statistics Theory and Methods*, 43, 1261–1276.
- Wilkinson, G. N., Eckert, S. R., Hancock, T. W., & Mayo, O. (1983). Nearest neighbour (NN) analysis of field experiments (with discussion). *Journal of the Royal Statistical Society B*, 45, 151–211.
- Williams, E. R., & Piepho, H. P. (2015). Optimality and contrasts in block designs with unequal treatment replication. *Australian and New Zealand Journal of Statistics*, 57, 203–209.
- Williams, E. R., & Piepho, H. P. (2018). An evaluation of error variance bias in spatial designs. *Journal of Agricultural, Biological and Environmental Statistics*, 23, 83–91.
- Williams, E. R., Piepho, H. P., & Whitaker, D. (2011). Augmented p-rep designs. *Biometrical Journal*, 53, 19–27.
- Williams, R. M. (1952). Experimental designs for serially correlated errors. *Biometrika*, 39, 151–159.
- Wolfinger, R. (1993). Covariance structure selection in general mixed models. *Communications in Statistics*, 22, 1079–1106.

SUPPORTING INFORMATION

Additional Supporting Information including source code to reproduce the results may be found online in the supporting information tab for this article.

How to cite this article: Piepho H-P, Michel V, Williams ER. Neighbor balance and evenness of distribution of treatment replications in row-column designs. *Biometrical Journal*. 2018;60:1172–1189. <https://doi.org/10.1002/bimj.201800013>

APPENDIX A

We here briefly describe the conditions for a neighbor-balanced design to exist with or without consideration of border plots. These conditions are mainly of theoretical interest for the types of trial we consider because more often than not the numbers of treatments, rows, and columns are such that only partial NB can be achieved at best. For this reason, the relevance of the difference between option (2), which ignores border plots, and the modification (2b), which takes border effects into account, is minor for the types of trials we consider, as both options can produce designs with partial NB. The choice does make a difference in cases where complete NB is attainable with one of the two approaches.

7	8	5	1	4	2	3	6	9	7	8
6	5	3	4	7	1	2	9	8	6	5
8	2	7	6	1	3	9	5	4	8	2
4	9	1	8	3	7	5	2	6	4	9

FIGURE A.1 Two-dimensional layout for $(v, k, s) = (9, 4, 9)$ generated using $(\gamma_R, \gamma_C, \gamma_E, \rho) = (1, 10^{-3}, 1, 10^{-1})$ [(option (2b)]. $\bar{f}_A^{RC} = 0.783$; $(n_0, n_1) = (0, 36)$; $MST_1 = 1.96$

A.1 Designs without consideration of border plots

In a row of s plots there are $s - 1$ nearest neighbors. For k rows, there is a total of $k(s - 1)$ adjacencies. There being $v(v - 1)/2$ pairs of treatments, it follows that

$$m = \frac{2k(s - 1)}{v(v - 1)}$$

must be a positive integer for a neighbor-balanced design to exist (Cheng, 1983; Kiefer & Wynn, 1981) and equals the number of adjacencies per pair. A further condition is that

$$r = \frac{ks}{v}$$

is a positive integer and equals the number of replications per treatment. One obvious setting where these conditions are fulfilled is when $k = s = v$ so that $m = 2$, which includes certain types of Latin squares (Kiefer & Wynn, 1981). As experimenters can not usually afford a high number of replications, attention can be restricted to the case $m = 1$. Two such options are $s = v$ and $k = v/2$ as well as $k = v$ and $s = (v + 1)/2$. For example, we may choose $s = v = 8$ and $k = 4$ (Example 2; Figure 9). The notion of NB considered in this section ignores the treatments that are planted on the plots bordering the trial area. It is common practice in the trials we are faced with to plant the same variety on border plots as on the neighboring plots on the margins of the trial area. Option (2) described in the main body of the paper targets this form of NB.

A.2 Designs with consideration of border plots

The field rows (blocks) can be thought of as derived from circular blocks on a ring that is, cut between two nearest neighbors and “unrolled.” The two neighbors will then be located at the two ends or the “unrolled” ring. The border plots would be allocated to the same treatments as those at the opposite ends of the inner block. Thus, if an unrolled block has treatment order

A C E D B,

we add treatment B at the left end and treatment A at the right end as border plots:

B | A C E D B | A (Azais et al., 1993).

On a ring with s plots, there are s adjacencies of nearest neighbors, so a design with k rows has ks adjacencies. This number must be divisible by $v(v - 1)/2$ for complete NB with border plots to exist, that is,

$$ks = m \frac{v(v - 1)}{2},$$

where m is the number of adjacencies per treatment pair (Lawless, 1971). The number of replications is $r = ks/v$, that is,

$$r = m \frac{(v - 1)}{2}.$$

It follows that the number of replications must be divisible by $(v - 1)/2$. Again, one will usually restrict attention to cases where $m = 1$. One convenient option is to choose $v = s$ and $k = (v - 1)/2$ or $v = 2k + 1$, for example, $(v, k, s) = (9, 4, 9)$ (Figure A.1). Alternatively, we may set $v = k$ and $s = (v - 1)/2$ or $v = 2s + 1$, e.g. $(v, k, s) = (9, 9, 4)$ (Rees, 1967).

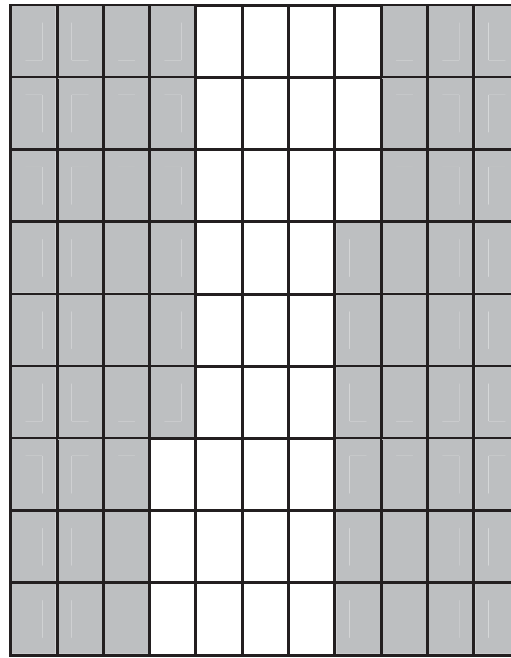


FIGURE B.1 A layout with $k = 9$ rows, $s = 11$ columns, and $g_s = 3$ *crooked* columngroups formed by grouping four, three, and four columns

Azais et al. (1993) distinguish left and right neighbors and define NB to require that left and right adjacencies must be the same for each treatment pair. This obviously requires twice the number of plots, that is, $r = m(v - 1)$, compared to when direction does not matter as we have assumed throughout the paper. Direction matters, for example, in polycross experiments with wind pollinated crops (Federer & Basford, 1991).

APPENDIX B

In Figure B.1 we describe *rectangular* blocks of unequal size formed by grouping columns (Example 1). Blocks were unequal in size because the number of column groups ($g_s = 3$) does not divide the number of columns ($s = 11$) in this example. An alternative in such cases is to form blocks as shown in Figure B.1. We call these blocks *crooked*. The first columngroup comprises the first three complete columns and the first six plots of column 4. The second superblock would comprise the remaining three plots of column 4, the complete columns 5, 6, and 7 and the first three plots of column 8. The third columngroup starts with the remaining three plots of column 8 and covers the complete columns 9–11 (Figure B.1). Note that the three columngroups in Figure B.1 are equal in size. Generally, if the columngroups are *crooked*, we can make them differ in size by at most one plot. Columngroups are formed by adding plots one by one, progressing down the columns. The process starts at the top left plot of the field layout, moving down the first column and adding plots until the column is exhausted. Then the path continues from the top of the second column, walking down in the same fashion, etc. The process is continued until the desired number of plots has been added. Then the next plot down the current column is the first plot of the next columngroup, and the same procedure is repeated until the second superblock is complete. The other superblocks are formed by continuing this procedure column by column until the whole layout is covered. In case of unequal columngroup sizes we first generate all larger blocks and then the smaller ones.